

EDA and MCMC Methods for Babies Dataset

Yatziri I. Carmona Ochoa

2026-04-26

```
library(UsingR)
library(Bolstad)
library(ggplot2)
library(reshape2)
library(ggribes)
```

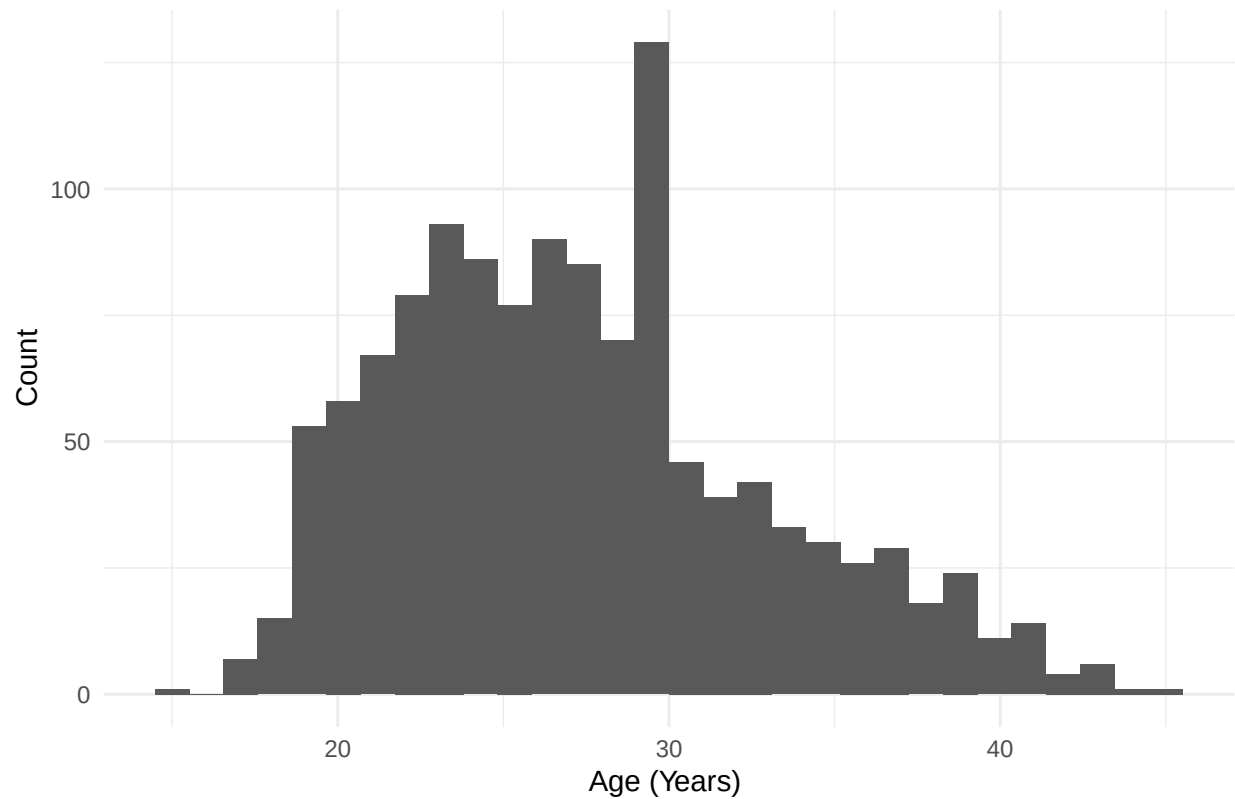
EDA

```
babies_clean <- babies

#removing missing/incomplete data
babies_clean[babies_clean == 999] <- NA
babies_clean$age[babies_clean$age == 99] <- NA
babies_clean$wt1[babies_clean$wt1 == 999] <- NA
babies_clean$smoke[babies_clean$smoke == 9] <- NA
babies_clean$race[babies_clean$race == 99] <- NA
babies_clean$inc[babies_clean$inc == 99] <- NA
babies_clean$inc[babies_clean$inc == 98] <- NA
babies_clean$sex[babies_clean$sex == 9] <- NA
babies_clean$ed[babies_clean$ed == 9] <- NA
babies_clean$ht[babies_clean$ht == 99] <- NA
babies_clean$parity[babies_clean$parity == 99] <- NA

#age distribution
ggplot(babies_clean, aes(x = age)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Mother's Age",
       x = "Age (Years) ", y = "Count") +
  theme_minimal()
```

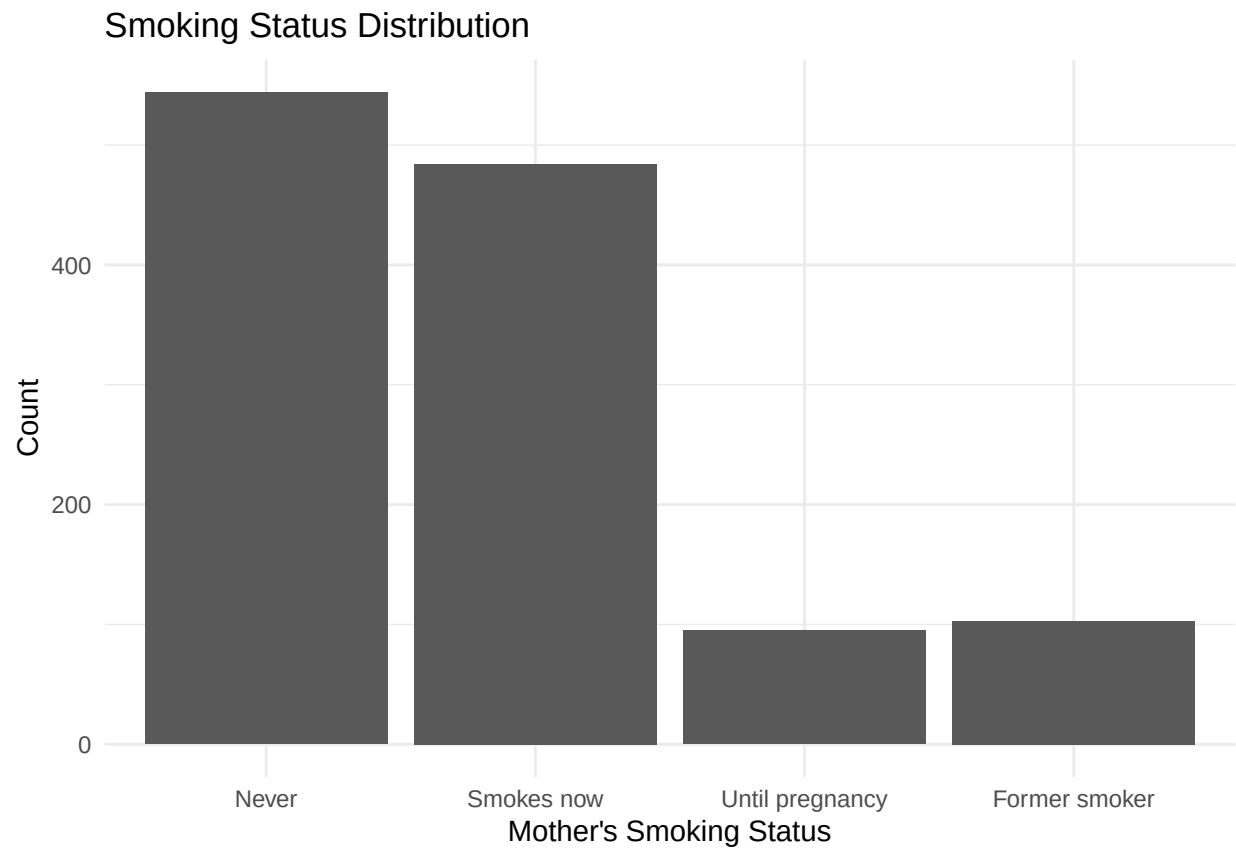
Distribution of Mother's Age



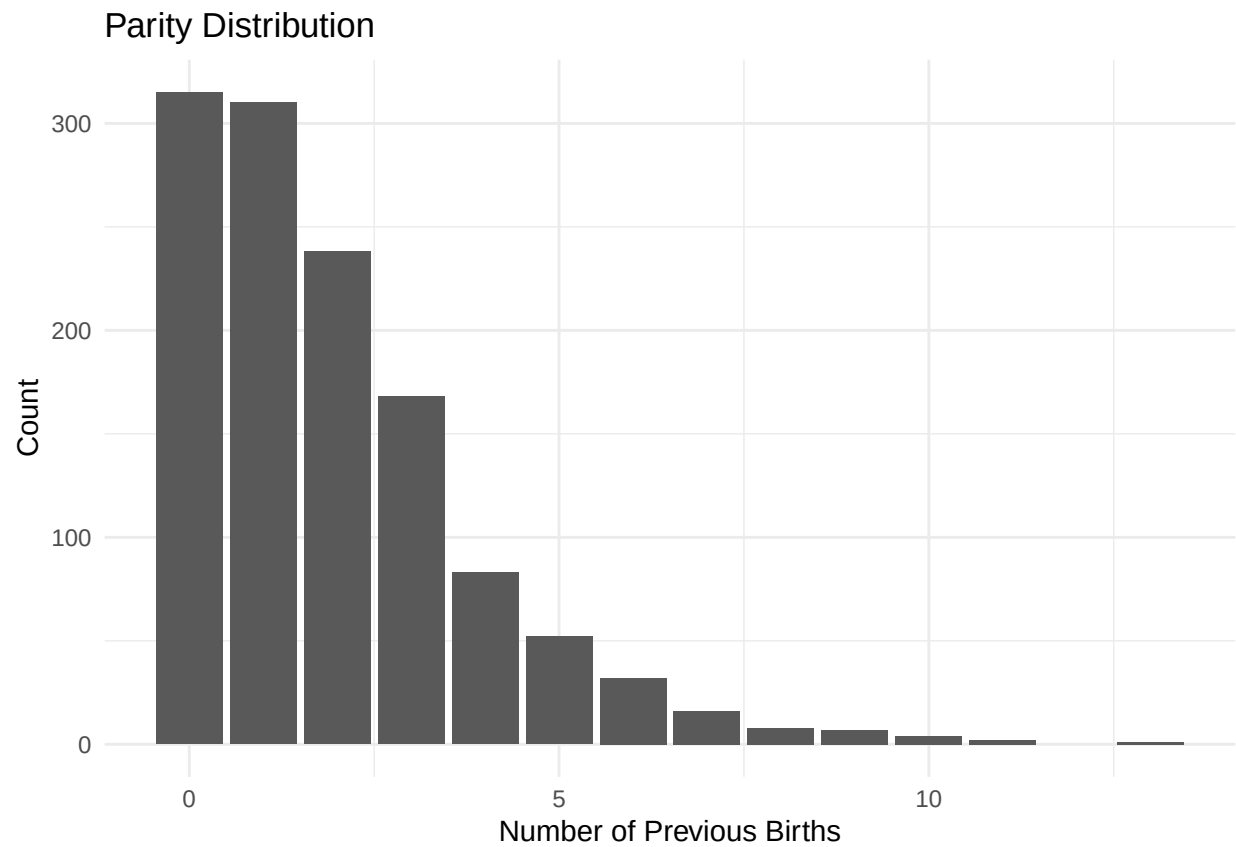
```
#smoking status distribution
babies_clean$smoke_group <- with(babies_clean,
  ifelse(smoke == 0, "Never",
    ifelse(smoke == 1, "Smokes now",
      ifelse(smoke == 2, "Until pregnancy",
        ifelse(smoke == 3, "Former smoker", NA))))))

babies_clean$smoke_group <- factor(
  babies_clean$smoke_group,
  levels = c("Never", "Smokes now", "Until pregnancy", "Former smoker")
)

ggplot(subset(babies_clean, !is.na(smoke_group)), aes(x = smoke_group)) +
  geom_bar() +
  labs(
    title = "Smoking Status Distribution",
    x = "Mother's Smoking Status",
    y = "Count"
  ) +
  theme_minimal()
```

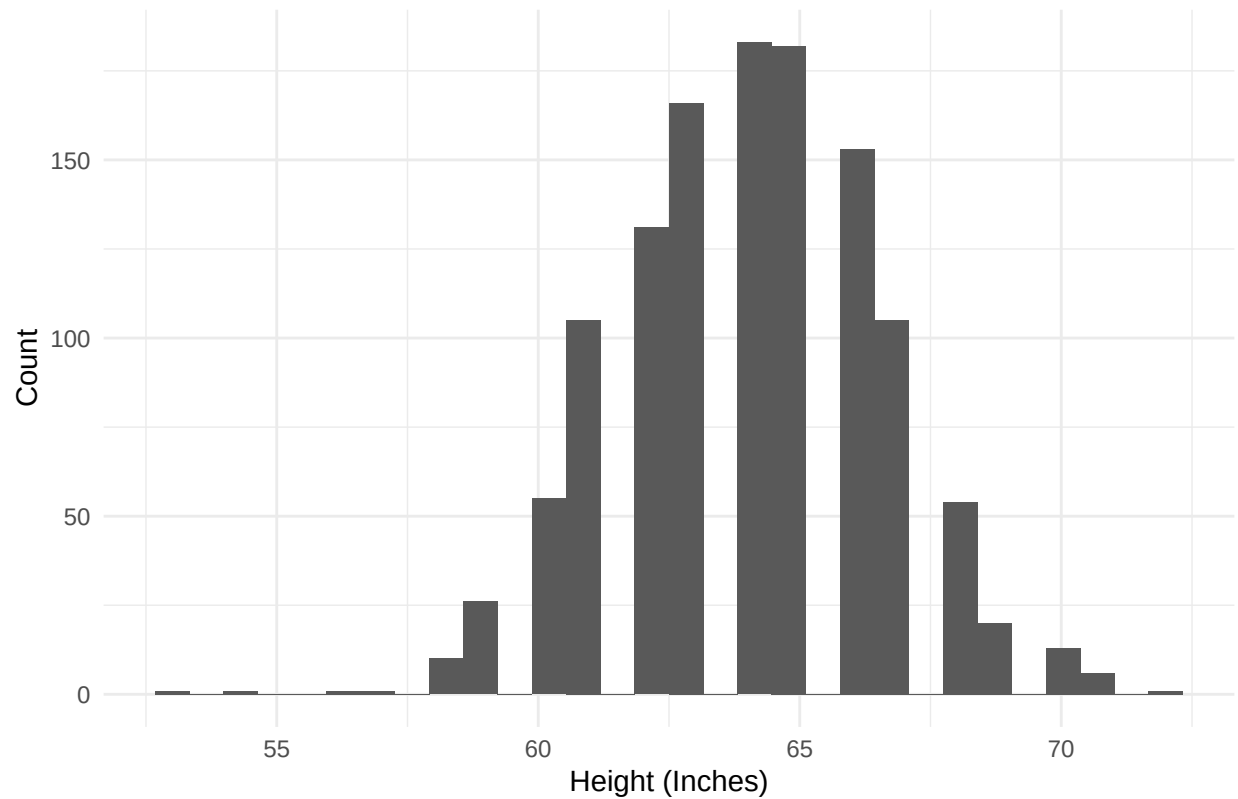


```
#parity distribution  
ggplot(babies_clean, aes(x = parity)) +  
  geom_bar() +  
  labs(title = "Parity Distribution",  
        x = "Number of Previous Births", y = "Count") +  
  theme_minimal()
```

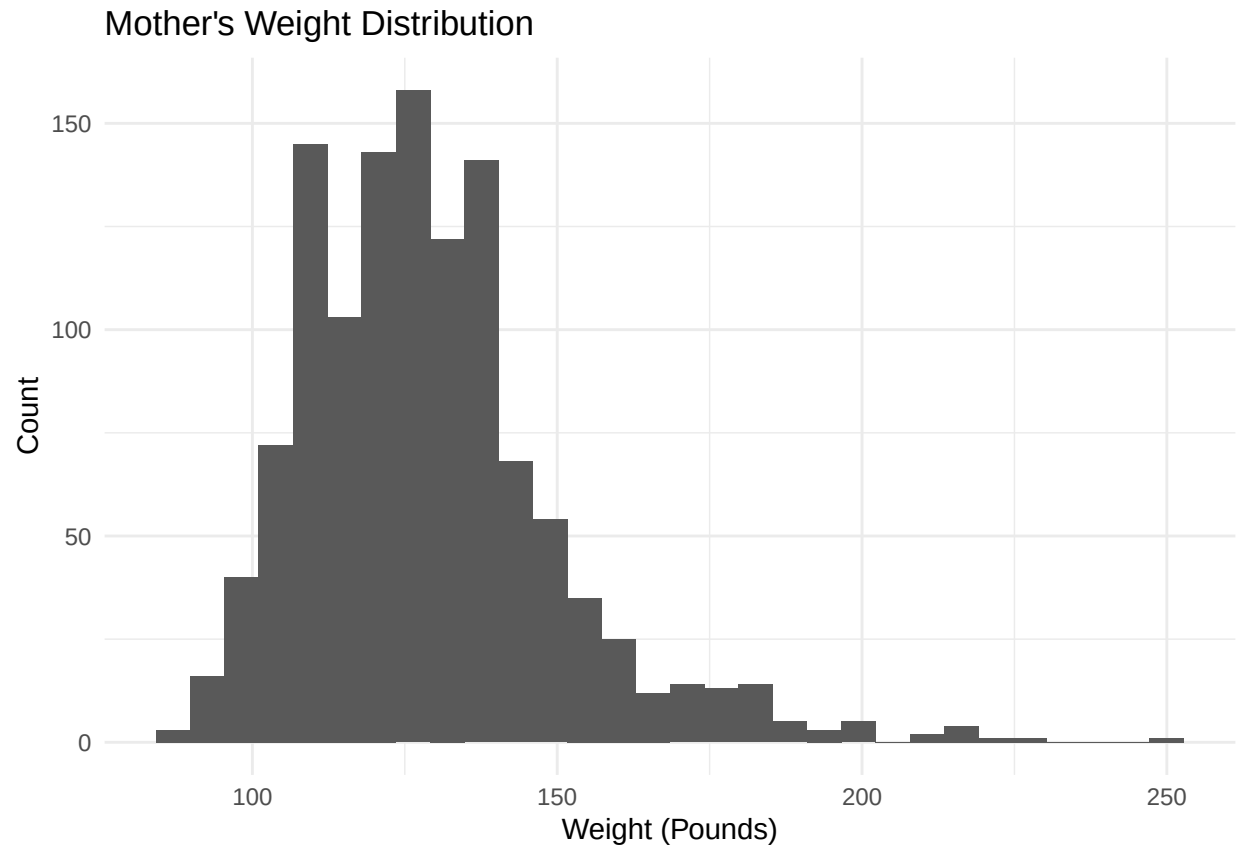


```
#height distribution for moms  
ggplot(babies_clean, aes(x = ht)) +  
  geom_histogram(bins = 30) +  
  labs(title = "Mother's Height Distribution",  
        x = "Height (Inches)", y = "Count") +  
  theme_minimal()
```

Mother's Height Distribution



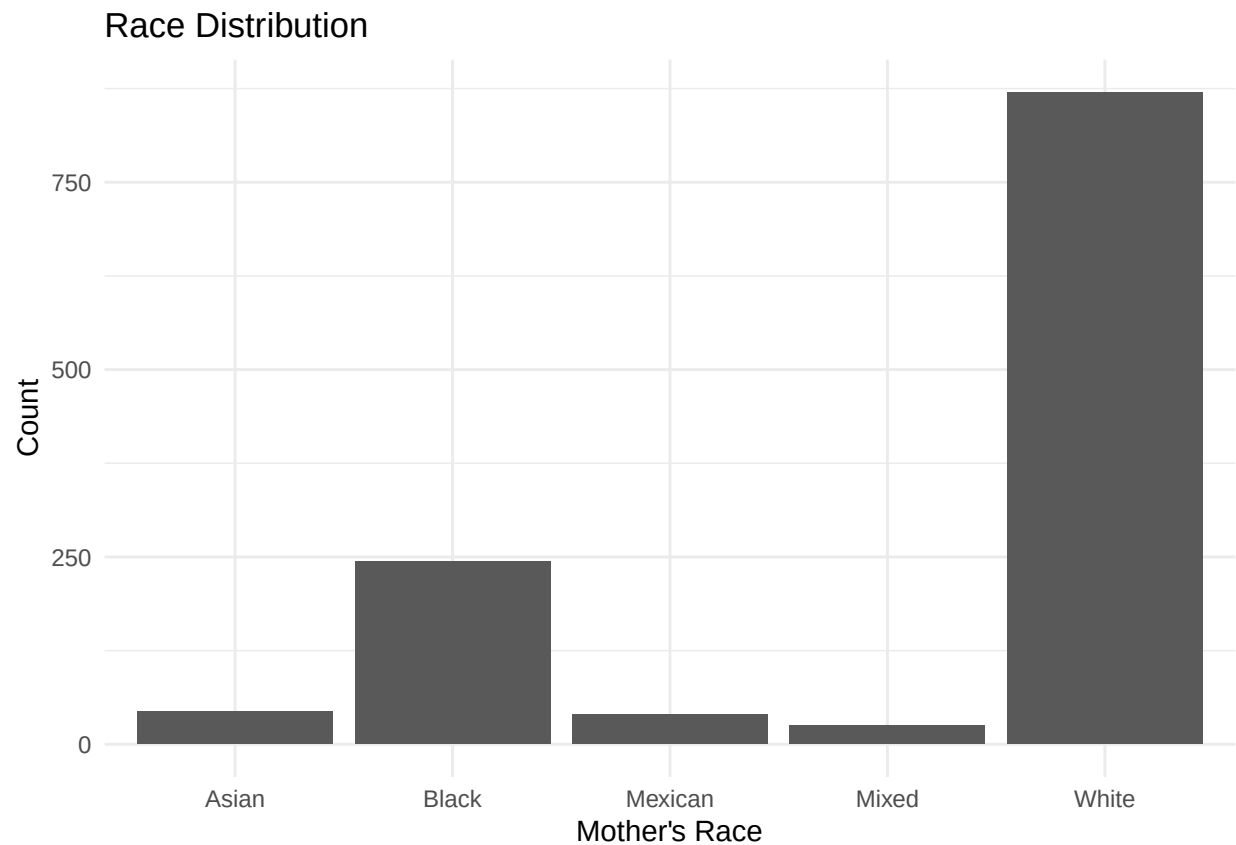
```
#mother's weight distribution  
ggplot(babies_clean, aes(x = wt1)) +  
  geom_histogram(bins = 30) +  
  labs(title = "Mother's Weight Distribution",  
        x = "Weight (Pounds)", y = "Count") +  
  theme_minimal()
```



```
#mother's race distribution
babies_clean$race_group <- with(babies_clean,
  ifelse(race %in% 0:5, "White",
    ifelse(race == 6, "Mexican",
      ifelse(race == 7, "Black",
        ifelse(race == 8, "Asian",
          ifelse(race == 9, "Mixed", NA))))))

babies_clean$race_group[babies_clean$race == 10] <- NA

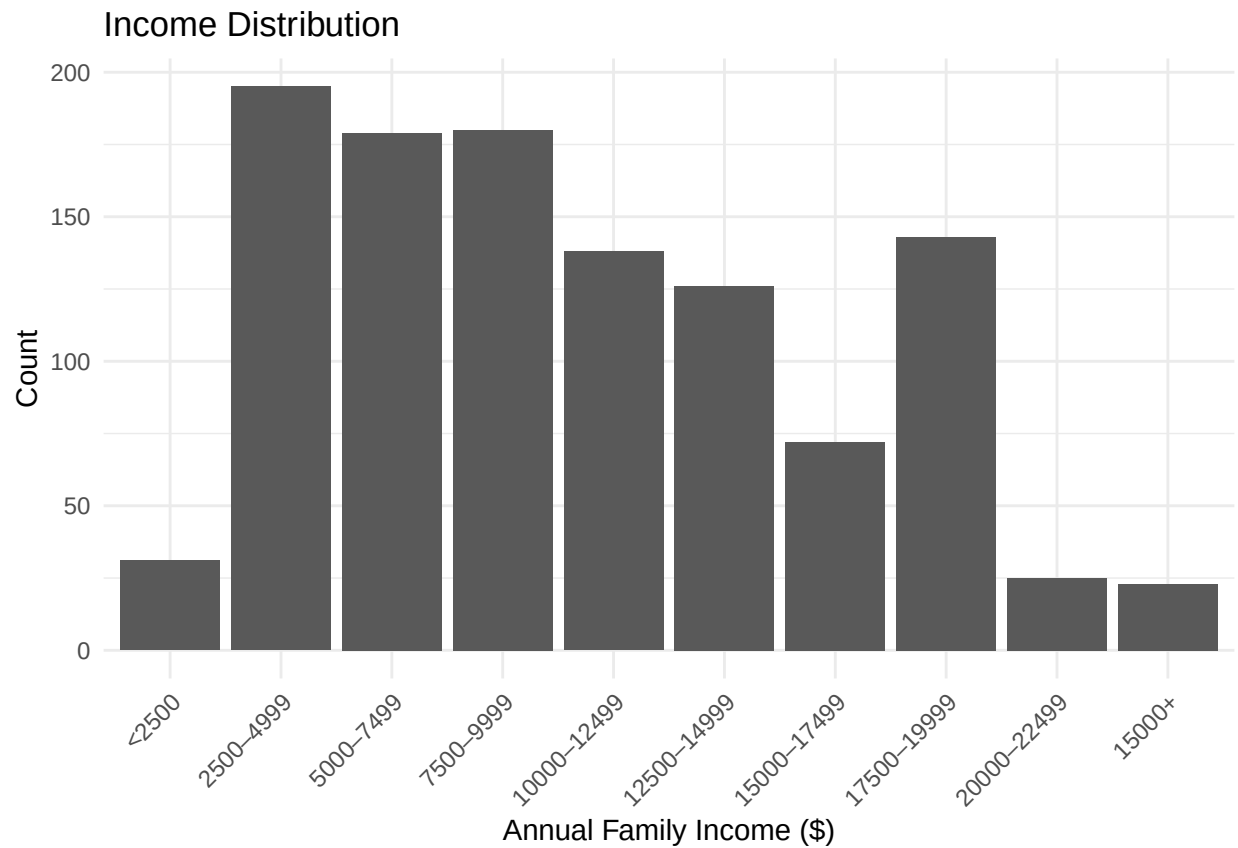
ggplot(subset(babies_clean, !is.na(race_group)), aes(x = race_group)) +
  geom_bar() +
  labs(
    title = "Race Distribution",
    x = "Mother's Race",
    y = "Count"
  ) +
  theme_minimal()
```



```
#annual income
babies_clean$inc_group <- factor(
  babies_clean$inc,
  levels = 0:9,
  labels = c(
    "<2500",
    "2500-4999",
    "5000-7499",
    "7500-9999",
    "10000-12499",
    "12500-14999",
    "15000-17499",
    "17500-19999",
    "20000-22499",
    "15000+"
  )
)

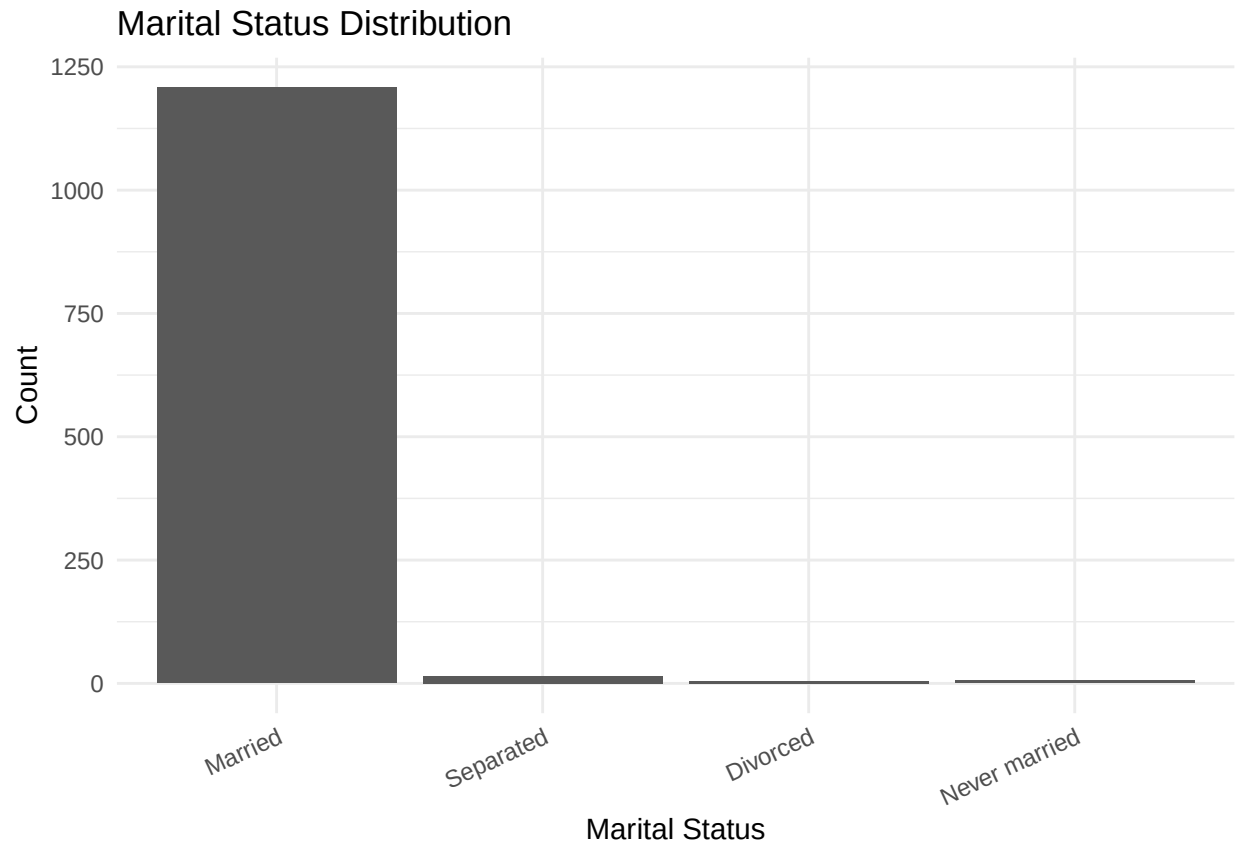
ggplot(subset(babies_clean, !is.na(inc_group)), aes(x = inc_group)) +
  geom_bar() +
  labs(
    title = "Income Distribution",
    x = "Annual Family Income ($)",
    y = "Count"
  ) +
  theme_minimal() +
```

```
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
#marital status distribution
babies_clean$marital_group <- factor(
  babies_clean$marital,
  levels = c(1,2,3,4,5),
  labels = c(
    "Married",
    "Separated",
    "Divorced",
    "Widowed",
    "Never married"
  )
)

ggplot(subset(babies_clean, !is.na(marital_group)), aes(x = marital_group)) +
  geom_bar() +
  labs(
    title = "Marital Status Distribution",
    x = "Marital Status",
    y = "Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 25, hjust = 1))
```

```
#mother's education distribution
babies_clean$edu_group <- with(babies_clean,
  ifelse(ed == 0, "Less than 8th grade",
  ifelse(ed == 1, "8th-12th (no diploma)",
  ifelse(ed == 2, "HS graduate",
  ifelse(ed == 3, "HS + trade",
  ifelse(ed == 4, "Some college",
  ifelse(ed == 5, "College graduate",
  ifelse(ed %in% c(6,7), "Trade school", NA)))))))))

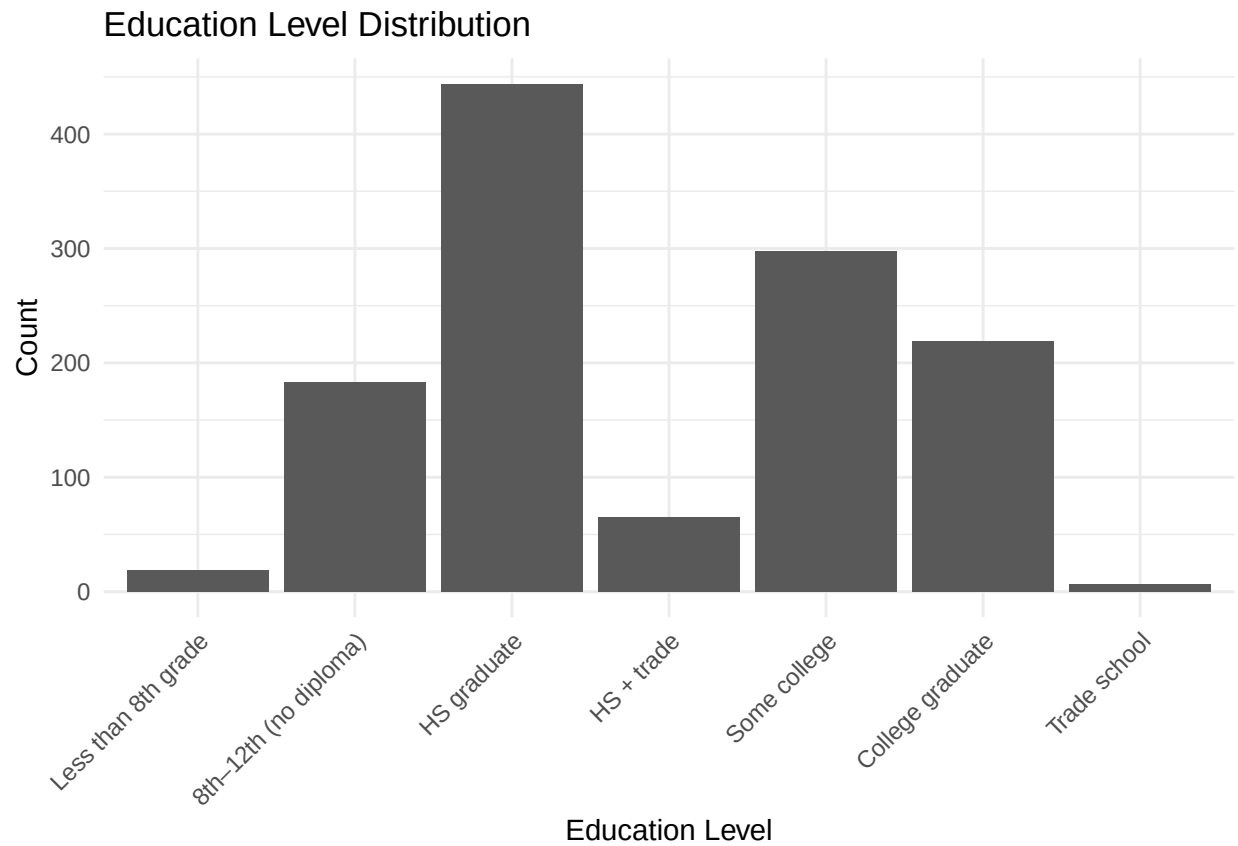
babies_clean$edu_group <- factor(
  babies_clean$edu_group,
  levels = c(
    "Less than 8th grade",
    "8th-12th (no diploma)",
    "HS graduate",
    "HS + trade",
    "Some college",
    "College graduate",
    "Trade school"
  )
)

ggplot(subset(babies_clean, !is.na(edu_group)), aes(x = edu_group)) +
  geom_bar() +
  labs(
```

```

title = "Education Level Distribution",
x = "Education Level",
y = "Count"
) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))

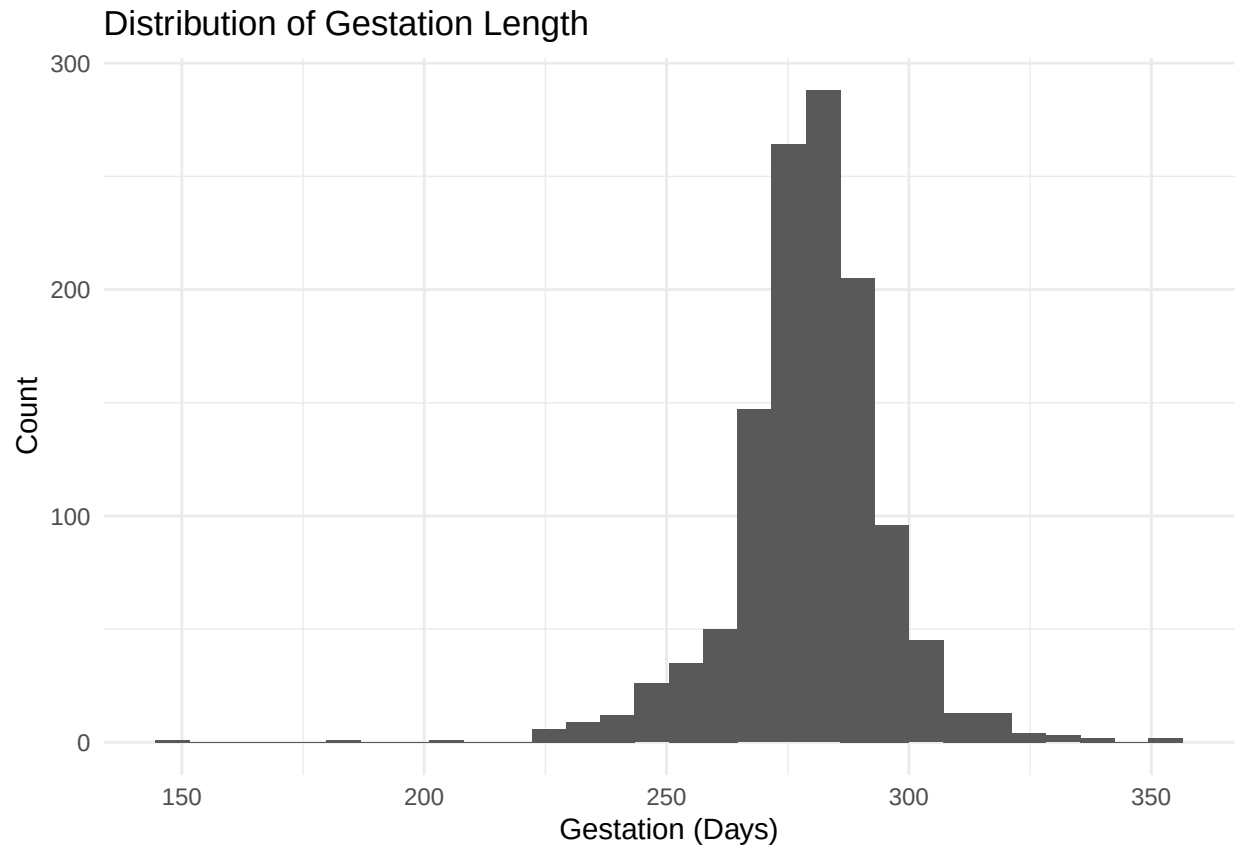
```



```

#gestation
ggplot(babies_clean, aes(x = gestation)) +
  geom_histogram(bins = 30) +
  labs(
    title = "Distribution of Gestation Length",
    x = "Gestation (Days)",
    y = "Count"
  ) +
  theme_minimal()

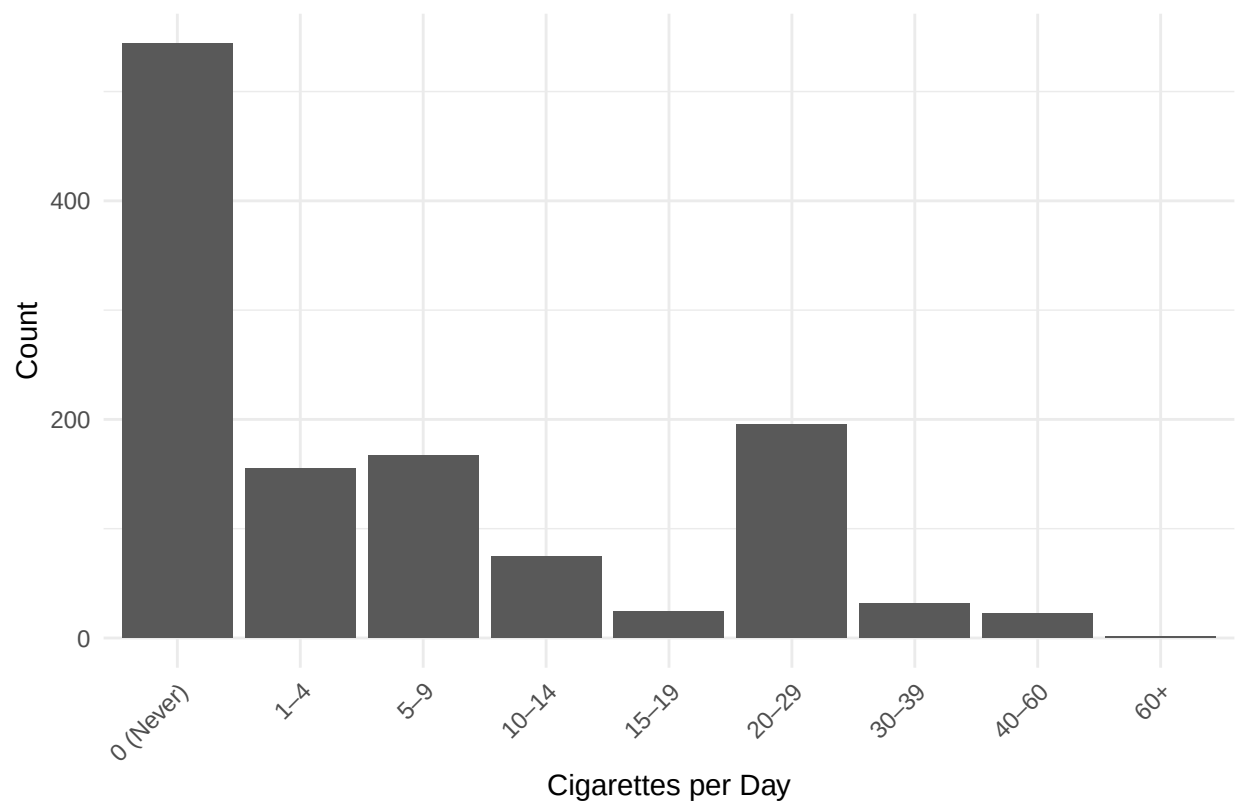
```



```
#number of cigarettes smoked
babies_clean$cigs_cat <- factor(
  babies_clean$number,
  levels = 0:8,
  labels = c(
    "0 (Never)",
    "1-4",
    "5-9",
    "10-14",
    "15-19",
    "20-29",
    "30-39",
    "40-60",
    "60+"
  )
)

ggplot(subset(babies_clean, !is.na(cigs_cat)), aes(x = cigs_cat)) +
  geom_bar() +
  labs(
    title = "Distribution of Cigarettes Smoked per Day",
    x = "Cigarettes per Day",
    y = "Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Distribution of Cigarettes Smoked per Day



MCMC for Bayesian Linear Regression

```
library(MCMCpack)

babies_df2 <- babies

#clean dataset
babies_clean_2 <- subset(babies_df2,
  wt != 999 &
  parity != 99 &
  age != 99 &
  ht != 99 &
  wt1 != 999 &
  smoke != 9 &
  !(number %in% c(9, 98, 99))
)

#create categorical smoking variable
babies_clean_2$cigs_cat <- factor(
  babies_clean_2$number,
  levels = 0:8,
  labels = c(
    "0 (Never)",
    "1-4",
```

```

    "5-9",
    "10-14",
    "15-19",
    "20-29",
    "30-39",
    "40-60",
    "60+"
  )
)

model <- MCMCregress(
  wt ~ gestation + ht + wt1 + number + parity + age,
  data = babies_clean_2,
  burnin = 2000,
  mcmc = 10000,
  thin = 5
)
summary(model)

##
## Iterations = 2001:11996
## Thinning interval = 5
## Number of chains = 1
## Sample size per chain = 2000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##              Mean          SD Naive SE Time-series SE
## (Intercept)  20.46387 13.847975 0.3096501      0.279934
## gestation    0.01113  0.006971 0.0001559      0.000161
## ht           1.41042  0.229107 0.0051230      0.004737
## wt1          0.05711  0.028310 0.0006330      0.000633
## number      -1.79584  0.248489 0.0055564      0.005556
## parity       0.06451  0.312991 0.0069987      0.006999
## age          0.03757  0.102709 0.0022966      0.002209
## sigma2      308.95856 12.962438 0.2898489      0.289849
##
## 2. Quantiles for each variable:
##
##              2.5%          25%          50%          75%          97.5%
## (Intercept) -5.504138 11.195160 20.19066 29.93518 49.55231
## gestation   -0.002068  0.006377  0.01084  0.01566  0.02513
## ht           0.955084  1.253830  1.41478  1.56675  1.85136
## wt1          0.003847  0.038243  0.05675  0.07518  0.11491
## number      -2.284053 -1.956842 -1.79795 -1.63088 -1.31222
## parity      -0.535379 -0.142989  0.05901  0.26994  0.69138
## age         -0.163235 -0.030945  0.03701  0.10590  0.23722
## sigma2      285.342717 300.116901 308.64709 317.22553 335.74883

#ridgeline plot
m <- as.matrix(model)
df <- as.data.frame(m)

```

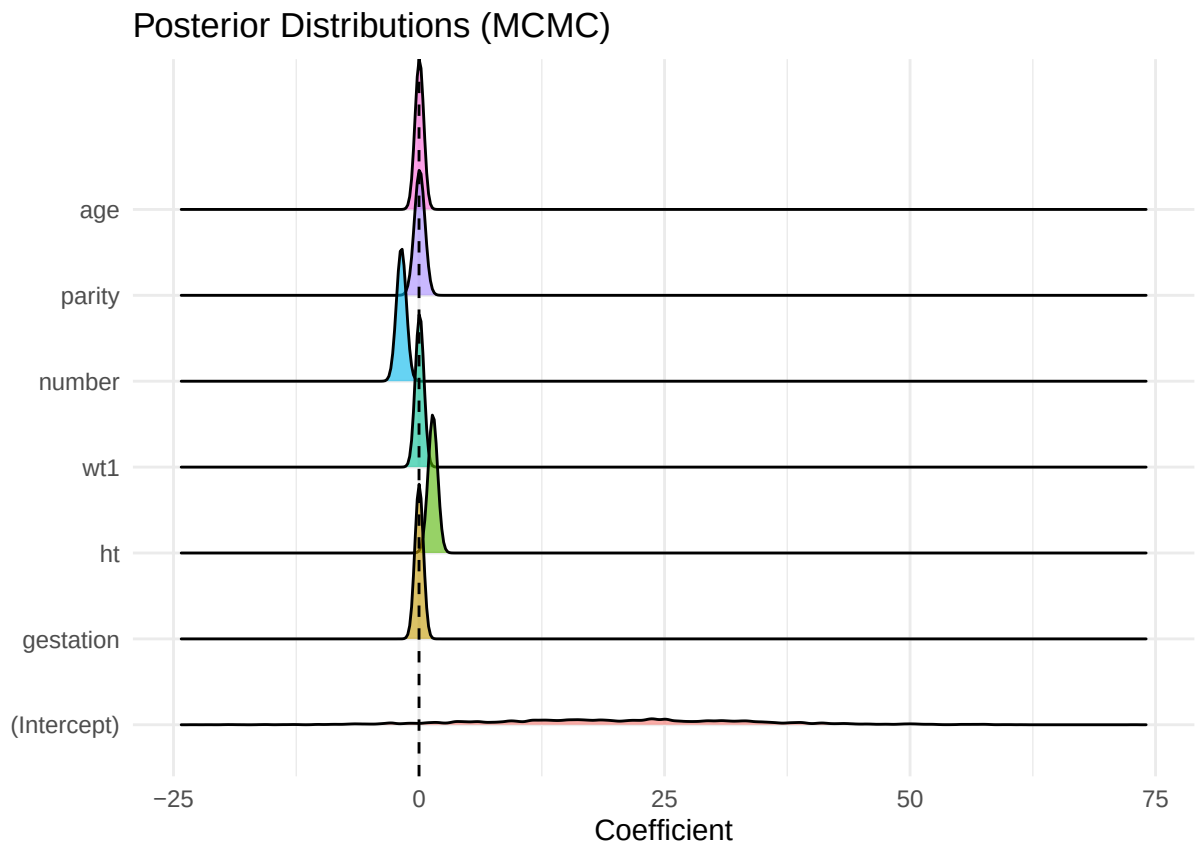
```

#remove variance parameter
df <- df[, !colnames(df) %in% "sigma2"]

#convert to long format
df_long <- melt(df)

#plot
ggplot(df_long, aes(x = value, y = variable, fill = variable)) +
  geom_density_ridges(alpha = 0.6) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Posterior Distributions (MCMC)",
    x = "Coefficient",
    y = ""
  ) +
  theme_minimal() +
  theme(legend.position = "none")

```



MCMC Informed Prior (Strong Means, Diffuse Variance)

```

b0 <- c(
  0.00, # intercept
  0.70, # gestation

```

```

0.84,    # ht
0.12,    # wt1
-1.80,   # number (cigs)
3.5,     # parity
0.00     # age
)

#precision matrix, required by MCMC pack
B0 <- diag(length(b0)) * (1 / 10000)

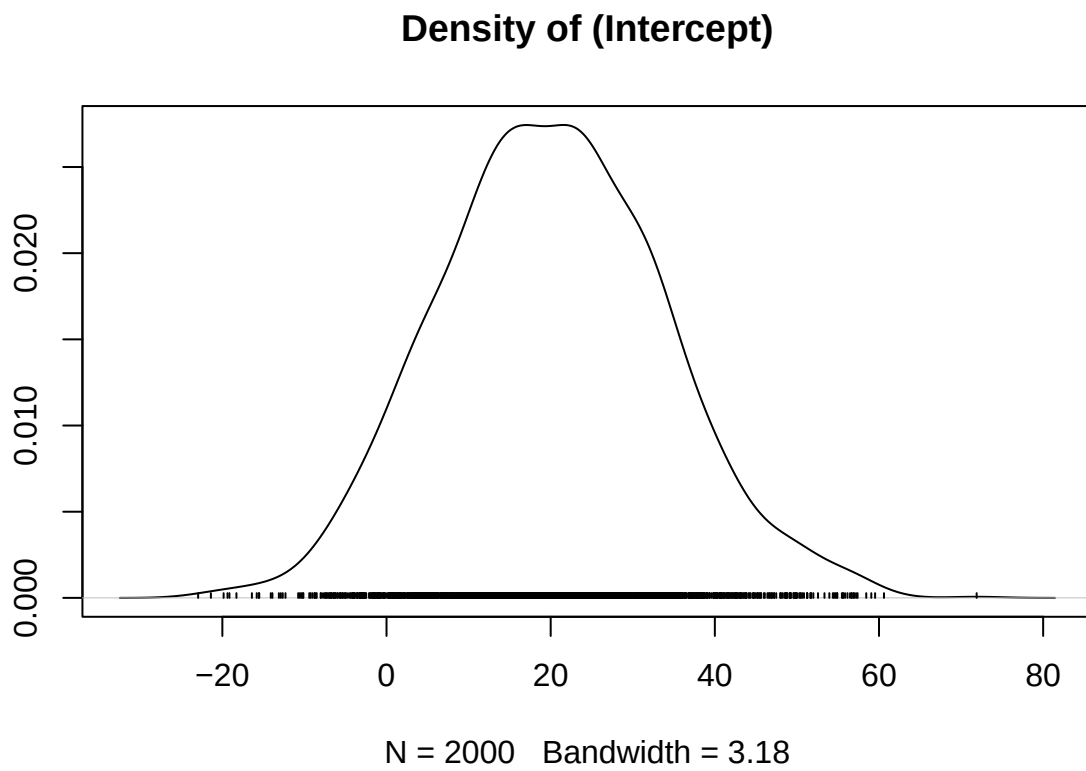
model_inf <- MCMCregress(
  wt ~ gestation + ht + wt1 + number + parity + age,
  data = babies_clean_2,
  b0 = b0,
  B0 = B0,
  burnin = 2000,
  mcmc = 10000,
  thin = 5
)

summary(model_inf)

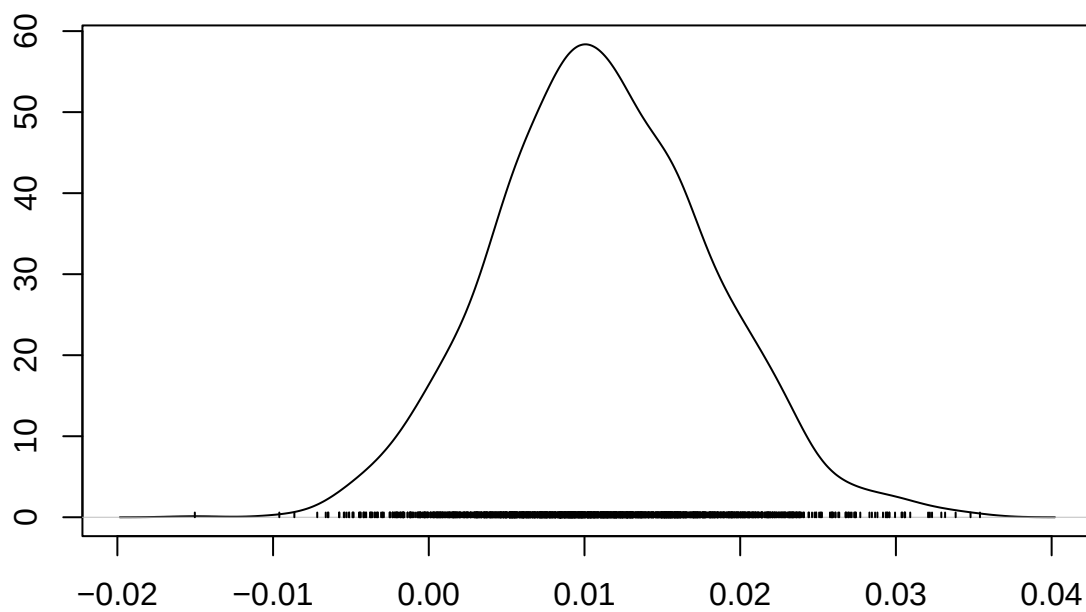
##
## Iterations = 2001:11996
## Thinning interval = 5
## Number of chains = 1
## Sample size per chain = 2000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##              Mean          SD Naive SE Time-series SE
## (Intercept)  20.08613 13.71998 0.3067880    0.2773522
## gestation    0.01115  0.00697 0.0001559    0.0001610
## ht           1.41638  0.22719 0.0050802    0.0046991
## wt1          0.05693  0.02829 0.0006327    0.0006327
## number      -1.79636  0.24848 0.0055561    0.0055561
## parity       0.06454  0.31299 0.0069986    0.0069986
## age          0.03808  0.10267 0.0022958    0.0022088
## sigma2      308.95440 12.96252 0.2898507    0.2898507
##
## 2. Quantiles for each variable:
##
##              2.5%          25%          50%          75%          97.5%
## (Intercept) -5.640166 10.906696 19.82416 29.46017 48.86751
## gestation   -0.002044  0.006396  0.01086  0.01568  0.02515
## ht          0.964447  1.260966  1.42091  1.57124  1.85349
## wt1         0.003765  0.038077  0.05658  0.07494  0.11476
## number      -2.284604 -1.957329 -1.79843 -1.63161 -1.31276
## parity      -0.535351 -0.142958  0.05904  0.26996  0.69140
## age         -0.162427 -0.030316  0.03739  0.10654  0.23756
## sigma2      285.334853 300.086056 308.63565 317.21774 335.73614

```

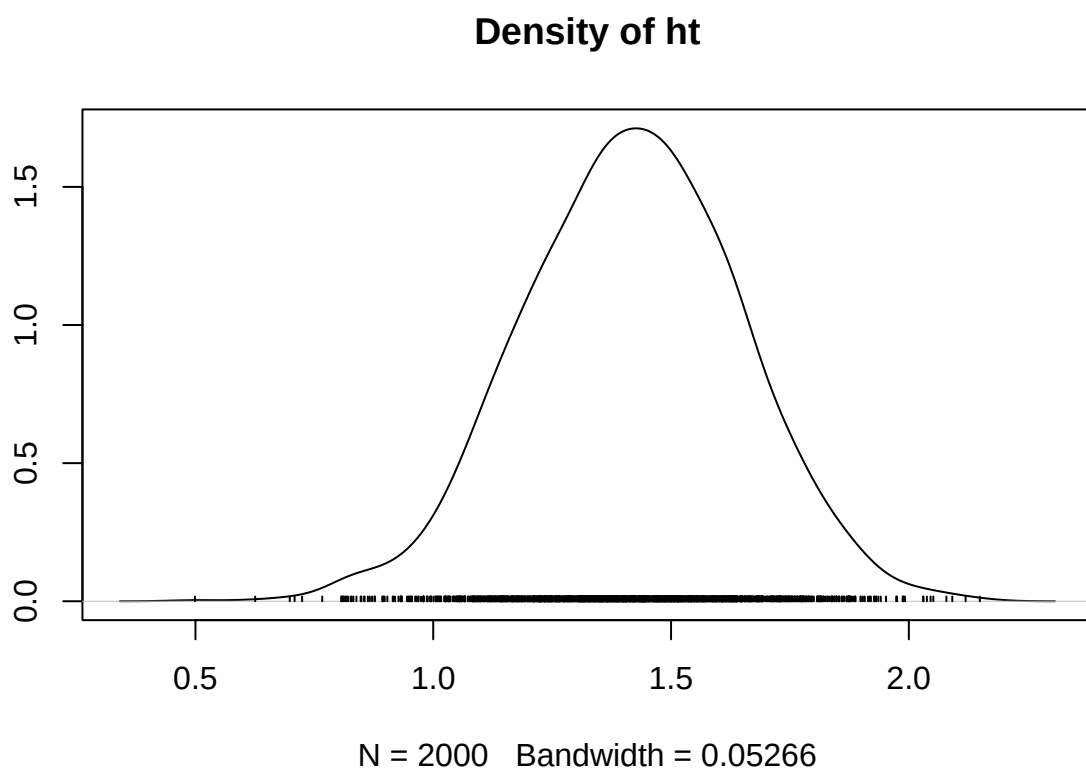
```
densplot(model_inf)
```



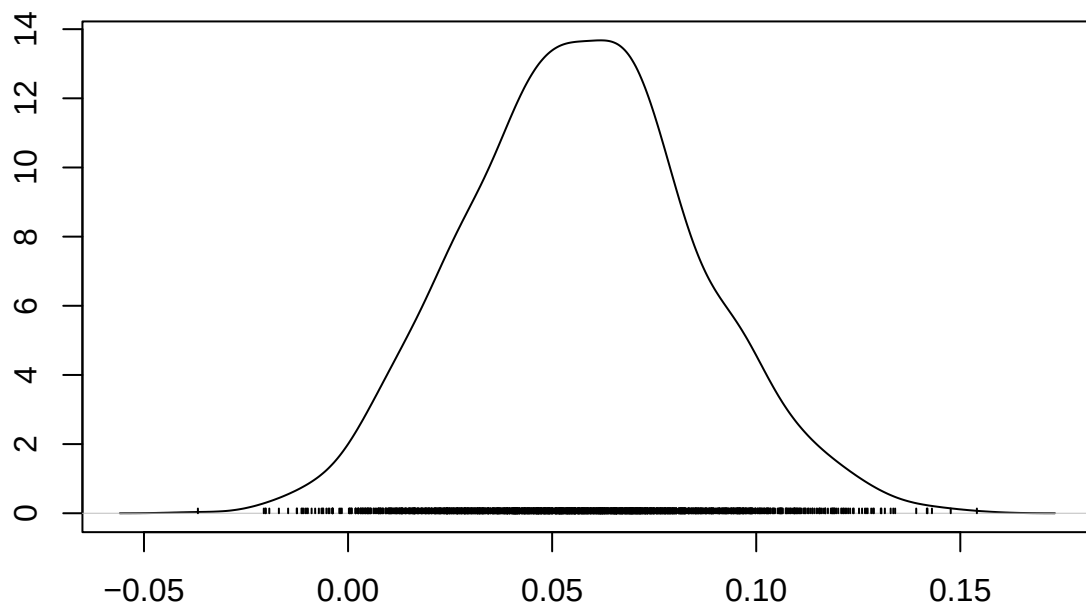
Density of gestation



N = 2000 Bandwidth = 0.001606

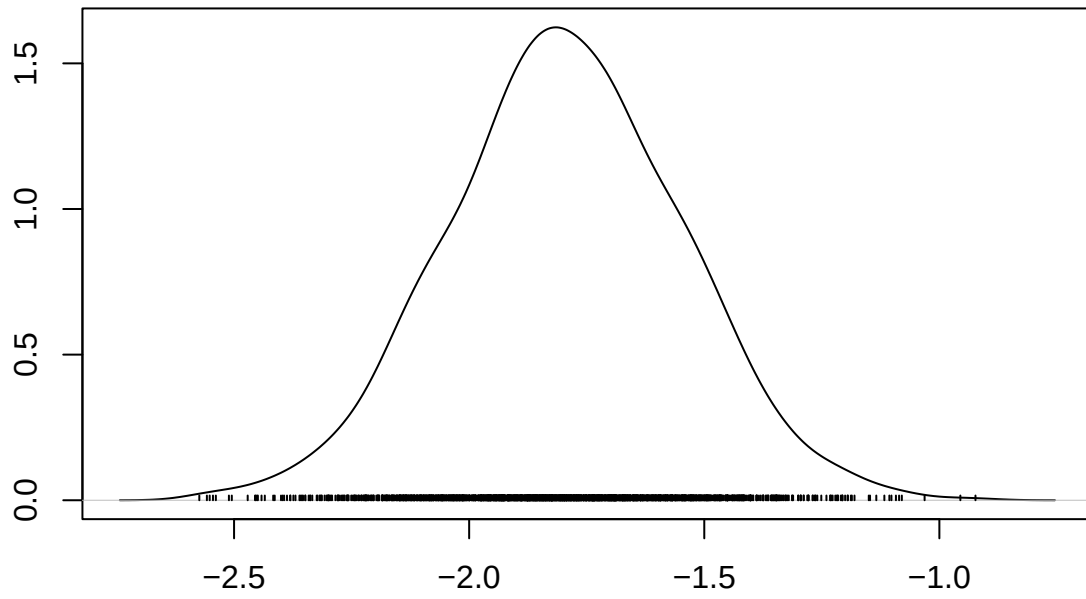


Density of wt1



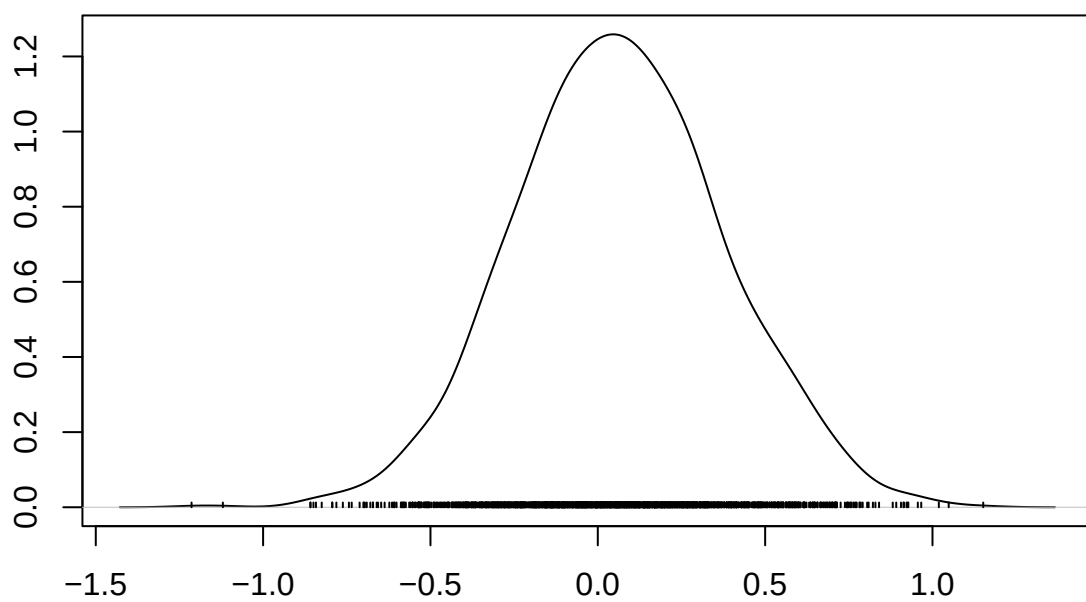
N = 2000 Bandwidth = 0.006376

Density of number



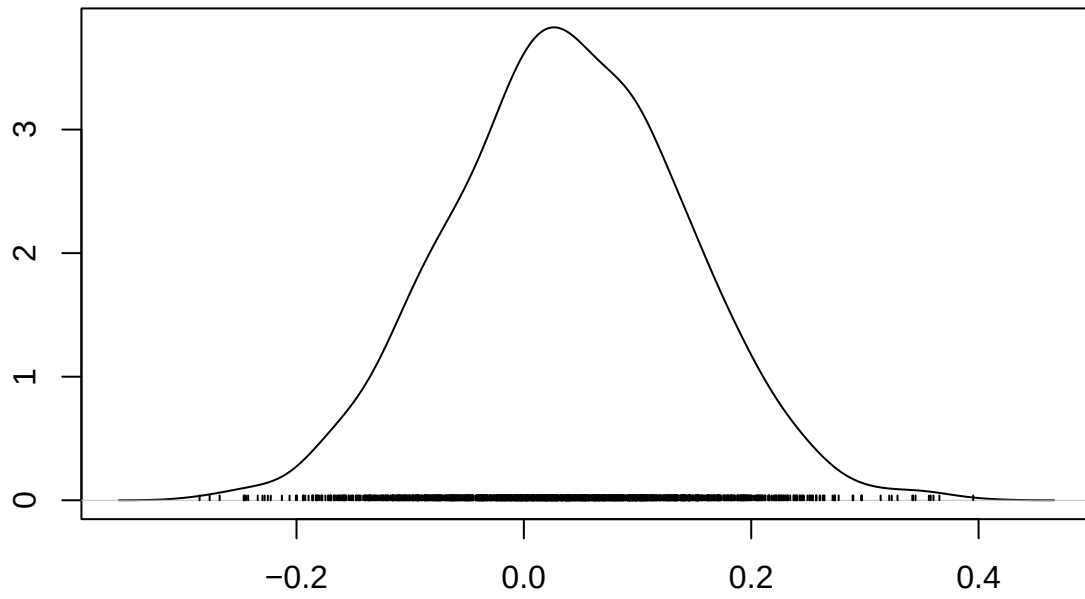
N = 2000 Bandwidth = 0.05634

Density of parity



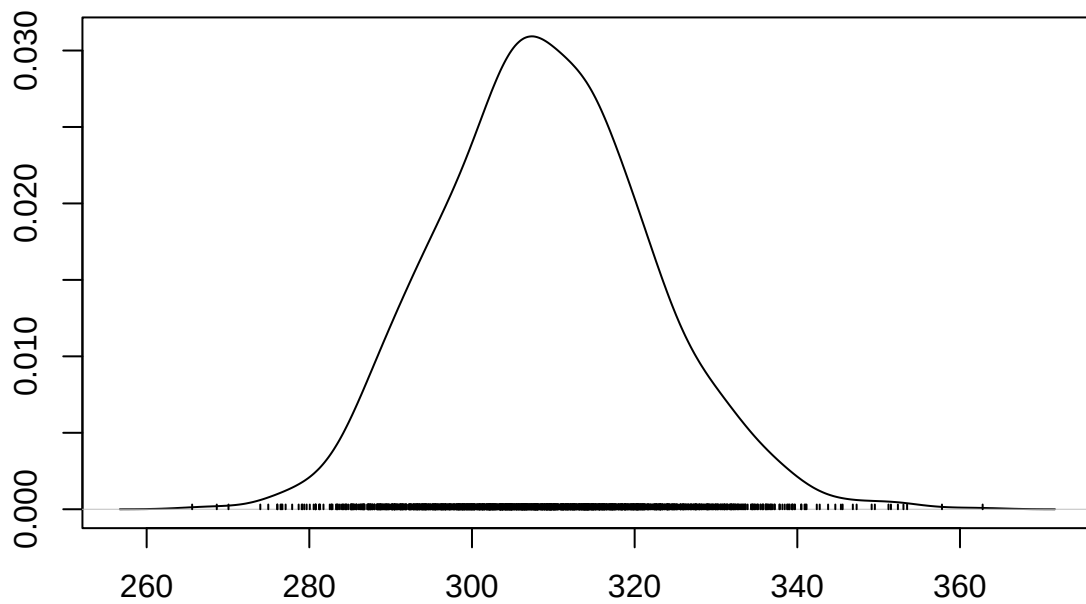
N = 2000 Bandwidth = 0.07143

Density of age



N = 2000 Bandwidth = 0.02367

Density of sigma2



N = 2000 Bandwidth = 2.963

```
#ridgeline plot
m2 <- as.matrix(model_inf)
df2 <- as.data.frame(m2)

#remove variance parameter
df2 <- df2[, !colnames(df2) %in% "sigma2"]

#convert to long format
df_long2 <- melt(df2)

#plot
ggplot(df_long2, aes(x = value, y = variable, fill = variable)) +
  geom_density_ridges(alpha = 0.6) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Posterior Distributions (MCMC)",
    x = "Coefficient",
    y = ""
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

